

Interview Data Capture for PTSD, Depression, and Anxiety Detection

Evidence base and scope

I could verify from the public OpenWillis repository that the project maintains a wiki with “research guidelines,” and that the library’s own facial-expressivity example uses both a target video and a baseline video as inputs for analysis. That is an important clue: in this ecosystem, facial expressivity is not treated as a purely absolute quantity; it is often interpreted relative to a within-subject reference condition. Because the two specific Brooklyn Health wiki pages you supplied were not directly retrievable from this environment, the concrete recommendations below are grounded in primary papers, public dataset documentation, and institutional preprints rather than the inaccessible page text itself. ¹

Across the literature, the strongest and most reusable examples come from a handful of interview corpora and clinical studies rather than from many small one-off experiments. The most informative sources for capture methodology are the DAIC-WOZ / Extended DAIC documentation, the Pittsburgh clinical depression interview study, AVID/AVEC challenge documentation, a veterans’ PTSD speech-biomarker study, the adolescent MMPSy anxiety/depression corpus, and a JMIR clinical interview paper that carefully documents downstream signal processing. Together, they show a fairly consistent pattern: researchers prefer close or separated microphones, at least roughly VGA-to-SVGA video, about 30 Hz video sampling, standardized interview prompts, and explicit post-collection cleaning rules for interruptions, silence, missingness, and tracking confidence. ²

What the available OpenWillis material confirms

The public repository confirms two things that matter for protocol design. First, OpenWillis explicitly frames its documentation as including “research guidelines,” not only software API notes. Second, the repository’s example for facial expressivity passes both a main interview video and a baseline video into the `facial_expressivity` function, which strongly suggests that standardized baseline capture is part of the intended analytic workflow. In practical terms, that aligns well with the depression and PTSD literature, where between-person variability in resting affect, head motion, and speaking style is substantial enough that uncontrolled capture can easily swamp clinically meaningful signal. ³

That baseline-oriented logic also fits the way established corpora are structured. In DAIC-WOZ, for example, the interview is semi-structured and standardized through a virtual interviewer, which reduces interviewer-induced variance and makes frame-level and utterance-level comparisons more interpretable. In the classic Pittsburgh depression study, the interview itself was a stable clinical instrument, the Hamilton Rating Scale for Depression, administered by interviewers trained to criterion. The larger methodological message is the same in both places: if the goal is biomarker extraction rather than ordinary clinical note-taking, the interview should be treated as a measurement protocol, not just a conversation. ⁴

Depression interview capture in the literature

The cleanest example of a tightly controlled depression interview capture pipeline is the Pittsburgh study by Cohn and colleagues. Participants were adults in a treatment trial for major depressive disorder, interviewed on as many as four occasions at roughly seven-week intervals using the Hamilton Rating Scale for Depression. Interviews averaged about 10 minutes. The capture setup used four hardware-synchronized analogue cameras and two microphones: two cameras recorded the participant's face and shoulders from approximately 15 degrees left and right, one camera recorded a full-body view of the participant, and one recorded the interviewer's face and shoulders from about 15 degrees to the interviewer's right. Video was digitized to 640×480 at 24-bit depth. The paper reports audio digitization in 10 ms analysis windows and used vocal prosody alongside facial action analysis. This is still one of the best documented examples of psychiatry-oriented multimodal interview capture. ⁵

A second major depression resource is DAIC-WOZ. Its interview sessions were designed to support detection of anxiety, depression, and PTSD, but the publicly documented depression package is especially detailed. The corpus contains 189 Wizard-of-Oz sessions lasting 7 to 33 minutes, with a 16-minute average. Audio was recorded from a head-mounted Sennheiser HSP 4-EW-3 microphone at 16 kHz. Each session package includes participant audio, transcripts, and face-derived features such as 2D landmarks, 3D landmarks, gaze, pose, action units, and HOG features extracted from an aligned 112×112 facial region. The documentation exposes frame-level confidence and `detection_success` flags, which are exactly the kind of low-level quality indicators a serious capture protocol should preserve. It also explicitly records technical problems: some sessions were excluded altogether, some contained interruptions, and some had incomplete interviewer transcripts. Identifiable speech segments were scrubbed by zeroing the waveform and corresponding feature entries. ⁶

A third depression-oriented example is the AViD/AVEC line of work, which is more task-based than clinician-interview-based but highly informative for capture specifications. The AViD-Corpus used in AVEC 2013 includes 340 video clips from 292 subjects, some recorded repeatedly at two-week intervals. Recordings were made in “quiet settings” using a webcam and a microphone. Tasks were PowerPoint guided and included sustained vowels, counting, read speech, singing, autobiographical positive and sad narratives, and Thematic Apperception Test style imagined stories. Audio was captured with a headset through a laptop sound card at 41 kHz and 16-bit depth. Original video files with mixed codecs and frame rates were resampled to a uniform 30 fps at 640×480 and 24-bit color, encoded as H.264 in MP4. For practical protocol design, AViD is useful because it shows that clinically relevant affective and depressive information can be collected with fairly ordinary hardware, provided the environment is quiet and the file format is standardized before analysis. ⁷

Not all depression studies stay at 640×480. A melancholia-focused facial behavior paper reported video capture at 800×600 and 30 fps using an AVT Pike camera, showing that some groups deliberately moved to slightly higher spatial resolution while keeping the now-standard 30 fps temporal sampling. The broader point is that 640×480 is workable and historically common, but 800×600 or higher can be justified if subtle facial movement is central and storage/bandwidth constraints are not severe. ⁸

PTSD and anxiety interview capture in the literature

For PTSD, the best documented interview-capture detail I found in a primary PTSD biomarker study is from Marmar and colleagues' work on speech-based markers in U.S. veterans. Their interviews were CAPS-based, and the audio of each interview was recorded in two channels using separate microphones for interviewer and participant. That small design choice matters a great deal. It reduces cross-speaker contamination, makes diarization much easier, and preserves speaker-specific acoustic measures that would otherwise be blurred in a single mixed track. Many later PTSD papers focus on model performance, but this study is a good reminder that clean channel separation is one of the most valuable upstream design decisions in speech biomarker work. ⁹

DAIC-WOZ and Extended DAIC also matter for PTSD because the corpus itself was designed around distress conditions including PTSD, not only depression. The corpus paper describes a standardized virtual interviewer, Ellie, and three development cycles: 120 face-to-face human interviews, 140 Wizard-of-Oz interviews, and a later automated system, with 351 participants recorded across two years. SimSensei's MultiSense perception framework synchronized audio and video and extracted real-time facial expression, gaze, head pose, smile intensity, speaking fraction, and speech dynamics. Even though the publicly accessible documents do not give a polished hardware bill of materials for every capture cycle, they make clear that PTSD-oriented interviewing in this line of work was conceived as synchronized multimodal sensing from the outset, not "audio-first with some extra video." ¹⁰

For anxiety, the literature is thinner when the requirement is specifically "interviewing patients" with detailed capture specs. One of the most concrete recent datasets is MMPsy, a large Mandarin adolescent corpus for anxiety and depression assessment. It is not a clinician-administered diagnostic interview and therefore should not be treated as directly equivalent to CAPS or HAMD-style interviewing, but it is a very instructive modern capture design. Data collection used a custom web application that administered 10 interview questions together with GAD-7 or PHQ-9 questionnaires. Audio responses were automatically recorded and uploaded. To assess response validity, the system embedded control questions. Preprocessing then removed inauthentic entries, mute recordings, recordings shorter than 1 second, and silent segments via voice activity detection; background noise was reduced with Spleeter; transcripts were generated automatically and manually verified for semantic consistency at the word level. After cleaning, the corpus retained 7,736 anxiety interviews and 4,247 depression interviews, with average durations around one minute. This is a strong example of how anxiety-oriented collection is currently being operationalized when collection must scale beyond the clinic. ¹¹

A related lesson comes from remote tele-video work. A 2024 multimodal tele-video study reported that features extracted from remote interviews could distinguish clinically diagnosed psychiatric disorder, major depressive disorder, self-rated depression, and self-rated anxiety. The accessible summary makes clear that the team extracted facial, vocal, linguistic, and cardiovascular patterns from synchronized remote audio-video, but it does not provide the kind of hardware-level detail that would let another group clone the exact capture environment. That gap is important in itself: in current anxiety and remote-psychiatry research, the analytic pipeline is often reported more fully than the camera or microphone specification. ¹²

Technical patterns that recur across trustworthy studies

The strongest cross-study pattern is that **30 fps is the de facto minimum normal target for facial video** in this domain. DAIC-derived facial features are framewise and confidence-tagged; AVID standardized video to 30 fps and 640×480; the JMIR clinical interview study processed facial action units at a 30 Hz video sampling rate; and the melancholia paper also used 30 fps. Researchers clearly treat this as sufficient for action-unit intensity, gross head motion, and conversational facial timing. There is far less support for lower frame rates when the goal is clinically meaningful facial movement analysis. ¹³

The second clear pattern is that **audio capture is usually made more robust than video capture**. In DAIC-WOZ, participants wore a head-mounted Sennheiser microphone at 16 kHz; in AVID, a headset microphone was used at 41 kHz and 16-bit; in the PTSD veteran study, each speaker had a separate microphone and separate channel. By contrast, exact camera models are frequently omitted. This suggests that researchers have learned, implicitly or explicitly, that audio contamination is more likely than facial-video failure to break the signal pipeline. If the budget forces a tradeoff, the literature argues for protecting microphone quality first. ¹⁴

The third pattern is that **room and protocol standardization matter more than expensive hardware**. AVID explicitly says recordings took place in quiet settings. DAIC-WOZ used a standardized virtual interviewer and carefully documented interruptions and excluded sessions with technical issues. MMPsy, which was collected in less controlled real-world settings, had to compensate later with noise removal, VAD, and control-question filtering. In other words, when the room is well controlled, the preprocessing burden goes down; when the room is uncontrolled, the post-processing burden rises sharply. ¹⁵

The fourth pattern is that **quality control is usually implemented as both capture-time design and post-hoc filtering**. DAIC-WOZ preserves tracking confidence, detection_success flags, interruption notes, scrubbed segments, and excluded-session lists. The JMIR clinical interview paper computed acoustic descriptors from 25 ms windows at a 10 ms step, sampled facial AUs at 30 Hz, then aligned both streams at 10 Hz, separated speaking and listening blocks, and even trimmed interview ends so unequal duration would not confound classification. MMPsy filtered out implausible responses and short or mute clips before modeling. Good psychiatric biomarker datasets do not just “record and analyze”; they preserve metadata that lets investigators identify when the recording should not be trusted. ¹⁶

The final recurring pattern is that **what gets measured is multimodal but not arbitrary**. Facial channels are usually action units, landmarks, pose, gaze, and some aggregate expressivity or motion statistics. Audio channels are usually prosody and voice quality features: pitch or F0, voiced/unvoiced decisions, harmonicity, spectral energy, formants, and more specialized measures such as NAQ, QOQ, H1H2, peakSlope, and Rd. The more reproducible studies also keep text or transcript information aligned to the acoustic stream. A practical collection protocol should therefore be designed backwards from those measurable endpoints rather than from vague ideas like “record a video and see what happens.” ¹⁷

Practical protocol and limitations

If your goal is to collect interview data that will work well with an OpenWillis-style facial-expressivity workflow and also remain comparable to the psychiatric ML literature, the most evidence-aligned baseline protocol is this:

- Use a **front-facing camera at 30 fps or higher**, and do not go below **640×480**; if bandwidth and storage allow, **800×600 or 720p** is a safer target for facial work, especially if you care about subtle upper-face and mouth dynamics. The literature consistently normalizes around 30 fps, with 640×480 common and 800×600 appearing in melancholia-focused work. ¹⁸
- Use a **close-talking headset or head-mounted microphone**; if interviewer speech matters analytically, record **separate channels** for participant and interviewer. This is much better supported than relying on a laptop mic in open air. ¹⁴
- Run the session in a **quiet room with minimal interruptions**, consistent seating, and stable face framing. The corpora that report “quiet settings” or tightly standardized interviews need less salvage preprocessing than the datasets gathered in more naturalistic settings. ¹⁹
- Build in **data quality checks as first-class outputs**: store frame-level face-tracking confidence, note interruptions, flag missing interviewer prompts, reject mute or ultra-short recordings, and preserve speaking-vs-listening segmentation for downstream features. Those practices are common in the strongest datasets and are directly useful later when model performance looks unstable. ²⁰
- Capture a **short neutral baseline clip** in addition to the main interview. That idea is consistent with the accessible OpenWillis example and with the general literature’s concern about individual differences in resting affect and motor style. ²¹

Several limitations remain important. Exact camera models, lensing, lighting geometry, distance to subject, and codec settings are often underreported even in otherwise strong psychiatric ML papers. The literature is much richer for depression and PTSD than for anxiety-specific clinician interview capture. Some of the most current anxiety datasets, like MMPsy, are scalable assessment interviews rather than formal clinician-administered diagnostic interviews. And for tele-video work, many reports describe what features were extracted while giving too little information about the actual consumer hardware, internet conditions, or room acoustics used during capture. Those omissions are not trivial; they are one of the main reasons why apparently similar models often fail to replicate across sites. ²²

¹ ³ ²¹ <https://github.com/bklynhlth/openwillis>

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² ¹³ ¹⁴ ¹⁵ ¹⁶ ¹⁷ ²⁰ https://dcapswoz.ict.usc.edu/wp-content/uploads/2022/02/DAICWOZDepression_Documentation.pdf

https://dcapswoz.ict.usc.edu/wp-content/uploads/2022/02/DAICWOZDepression_Documentation.pdf

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⁵ <https://sites.pitt.edu/~jeffcohn/biblio/Cohn%20ACII2009.pdf>

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